

DR. VASUDEVA KUMAR MANTRAVADI

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Indian national | Available for UAE / KSA / Qatar / Oman relocation | 60 days notice

DYNAMIC SIMULATION · DIGITAL TWIN · OPERATOR TRAINING SIMULATORS — UPSTREAM OIL & GAS

PROFILE

Process engineer with 20 years delivering dynamic simulation, operator training simulators and real-time production systems for upstream operators, onshore fields and FPSOs. PhD, IIT Madras; postdoctoral research, Max Planck Institute. Deep working ownership of K-Spice, LedaFlow, OLGA and Aspen HYSYS across the full asset lifecycle — design verification, commissioning, start-up support and live operations. Recent work applies machine learning and GenAI to production surveillance, artificial lift diagnostics and gas lift optimisation on operating assets. Author of Integrated Process Engineering: Standards, Simulation, Process Safety and Digital Twins (2026).

CORE COMPETENCIES

Simulation & Modelling: K-Spice, LedaFlow, OLGA, Aspen HYSYS, Aspen Plus, UniSim, gPROMS, DySim — steady-state, dynamic and transient multiphase

Digital Twin & OTS: Operator Training Simulator design and delivery, DCS/control-scheme testing, model validation against plant data, production system simulators, real-time model integration

Flow Assurance & Production: Hydrate management and inhibition curves (Multiflash), nodal analysis (IPR/VLP), hydraulics, compressor anti-surge, gas lift and artificial lift optimisation, production surveillance

AI / ML for Operations: Failure prediction for ESP, PCP, gas lift and SRP systems; feature engineering from historian data; LLM-based engineering assistants; Python, OSI PI, Kafka, JSON interfaces

Process Safety & Assurance: HAZOP facilitation, ISO 17776, design verification against FEED/EPC basis, technical assurance, asset integrity

Leadership: Multidisciplinary team leadership, client technical authority, pre-sales solutioning and project costing, mentoring, training design and delivery

SELECTED PROJECT DELIVERY

- ONGC Ankleshwar — architected and deployed the real-time production monitoring and optimisation system covering 320+ wells across 26 facilities.
- Reliance KGD6 (R-Cluster, SAT-Cluster) and MJ FPSO — delivered Production System Simulators supporting first start-up, operator readiness and ongoing production monitoring for deepwater offshore assets.
- Equinor Snøhvit Future Phase 2 — conducted FEED/EPC design verification using dynamic simulation for new compressor installation, providing the technical basis for design modification.
- ONGC Nandasan — delivered the Production Management System for an onshore field, integrating well models with facility-level constraints.
- Shell global portfolio — led deployment of enterprise simulation tooling (AspenOne, OLGA) across 18 upstream, refining and chemicals sites, supporting production and margin decisions on multi-billion-dollar assets.

PROFESSIONAL EXPERIENCE

Principal Process Engineer

FALKOR (formerly Kongsberg Digital India Software & Solutions) | Bangalore, India | Apr 2020 – Present

- Technical lead for end-to-end delivery of engineering, real-time and Operator Training Simulator solutions to upstream operators — owning solution architecture, model build, factory and site acceptance testing, and commissioning support.
- Delivered real-time production systems and production system simulators for ONGC and Reliance assets, including start-up support for the MJ FPSO.
- Built an end-to-end LedaFlow–K-Spice coupled simulator with live data exchange, performing sucker rod pump calculations and streaming results to the operations visualisation layer.
- Developed K-Spice Tutor, an LLM-based engineering assistant that guides process engineers through simulator workflows, reducing onboarding time for new users.
- Delivered machine-learning failure prediction for artificial lift systems (ESP, PCP, gas lift, SRP), including engineered feature sets and failure signature matrices used in client well surveillance workshops.
- Led the winning team at AI Hackathon 2025 (hybrid GenAI/physics gas lift optimisation for mature wells) and at AIVOLUTION 2024 (hybrid AI gas lift optimisation).
- Facilitated HAZOP studies for major production assets to ISO 17776, and provided design verification against FEED/EPC basis for brownfield modifications.
- Directed delivery of five simulation and production system deployments over 24 months, leading a team of [X] engineers across [X] client accounts.

- Provided technical solutioning and project costing support to regional sales, contributing to bid submissions and client acquisition.

Process Engineering Specialist

Shell Technology Centre Bangalore | Bangalore, India | Mar 2015 – Mar 2020

- Managed a multi-year, multi-million-dollar upstream simulation tools programme in collaboration with Flow Assurance, SLB and Process Engineering teams.
- Led global deployment of the AspenOne Engineering Suite across Shell chemicals and refining sites, including engineer training delivery in Singapore.
- Built dynamic simulations of DCS control schemes for commissioning readiness, materially reducing commissioning duration and improving start-up stability.
- Delivered process engineering solutions for asset integrity programmes across 18 global sites using AspenOne and OLGA.

Scientist — Control & Optimisation

ABB Global Industries and Services | Bangalore, India | Sep 2011 – Mar 2015

- Developed performance monitoring and techno-commercial analysis for upstream oil and gas assets, including a UniSim-based prototype.
- Delivered optimisation pilots for large-scale RO desalination plants in Jeddah, Gujarat and the UAE, validating measurable productivity and energy savings.
- Authored three patent filings in renewable energy integration and model-based optimisation.

Scientist / Engineer

Procter & Gamble | Bangalore, India | Jan 2010 – Aug 2011

- Built a flow model library and a pneumatic conveying modelling platform for solids and non-Newtonian fluids, deployed to process engineers across global manufacturing sites.
- Led process design optimisation across 12+ manufacturing sites using Aspen HYSYS and IDEAS.

Process Simulation Engineer II

Invensys India Development Centre | Hyderabad, India | Feb 2009 – Jan 2010

- Developed valve flashing calculation features for DySim dynamic simulation software and resolved priority technical issues raised by ExxonMobil.
- Validated DySim simulation output against plant data, identifying model improvements and strengthening client confidence in results.

PUBLICATION

Mantravadi, V. K. Integrated Process Engineering: Standards, Simulation, Process Safety and Digital Twins from Design to Operations. 2026. A practitioner reference covering ASME, API, ISO, NFPA, IEC, ISA, TEMA and BS/EN codes alongside dynamic simulation and digital twin workflows across the asset lifecycle.

Seven peer-reviewed journal and international conference publications in process modelling, nonlinear analysis and optimisation. Three patent filings.

EDUCATION

Postdoctoral Fellow, Chemical Engineering: Max Planck Institute for Dynamics of Complex Technical Systems, Magdeburg, Germany — 2009. Iron oxide kinetics for hydrogen purification in fuel cell applications.

PhD, Chemical Engineering: Indian Institute of Technology Madras — 2009. CGPA 9.35/10. Modelling and nonlinear analysis of chemical and biological reactor-separator systems.

M.Tech, Chemical Engineering: National Institute of Technology (formerly REC) Warangal — 2002. Computer-Aided Process Equipment Design. First in batch.

B.Tech, Chemical Engineering: Acharya Nagarjuna University, Guntur — 2000. Second in university.

CERTIFICATIONS & RECOGNITION

- BOSIET with Emergency Breathing System (offshore survival) — 2022
- Winner, AIVOLUTION 2024 — hybrid AI gas lift optimisation